

Familiarisation with TPS



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1 Objective

1. Create a radiotherapy treatment plan using a computerised TPS (Treatment Planning System).
2. Perform treatment planning using various modes (e.g., photons, protons, etc.) of treatment and using different setups of beams to find the best treatment plan.

2 Apparatus

- A computer with MatRad software.
- DICOM image of the Patient.

matRad is an open-source treatment planning system (TPS) research and education toolkit developed in MATLAB for external beam radiotherapy planning [1]. It supports the complete basic planning workflow, including DICOM data import, structure handling, dose calculation, inverse planning, and plan evaluation in a graphical user interface. A major strength of matRad is that it allows planning for different modalities such as photons, scanned protons, and scanned carbon ions within a common framework, which makes it useful for comparative learning.

In this experiment, matRad was used as the primary software platform to understand TPS operation, starting from patient image loading to beam setup and optimization. The software enables users to visualize target volume coverage and dose to organs at risk (OARs), and to assess plan quality through dose distributions and dose-volume histograms (DVHs). Because the code is open and modular, it is particularly suitable for academic use, where both clinical concepts and underlying computational methods can be studied together [1].

3 Theory

The most common type of radiation therapy is external beam therapy, which is characterised by directing ionizing radiation from the outside into the patient's body. Treatment planning is an integral part of radiation therapy treatment, performed before the actual patient treatment using specialized software. The treatment planning process aims to simulate the dose deposition within the patient's body and to optimize the dose distribution according to clinical objectives and constraints that define a trade-off between tumour irradiation and normal tissue sparing.

3.1 Treatment Planning Process

The treatment planning process involves several steps, which are outlined below in Figure 1.

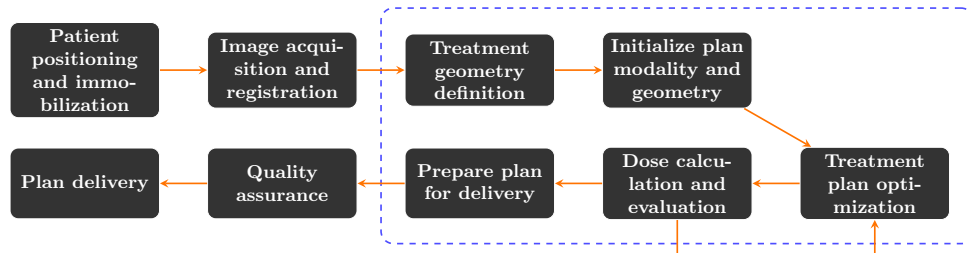


Figure 1: The treatment planning workflow. The dotted blue region represents the TPS-internal steps covered in this experiment.

3.1.1 Treatment Plan Prescribing and reporting

In external photon beam radiotherapy, prescription and reporting require clear definition of target-related volumes and dose-reporting quantities so that treatment intent is unambiguous and reproducible [2].

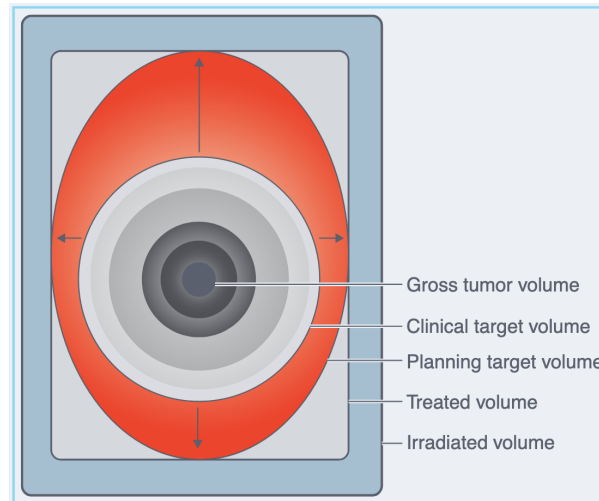


Figure 2: ICRU target and dose specification concepts used in treatment plan prescribing and reporting.

- **Gross Tumor Volume (GTV):** The gross demonstrable extent of malignant disease identified clinically or through imaging.
- **Clinical Target Volume (CTV):** GTV plus regions suspected to contain microscopic disease; it represents the true tissue volume that must receive adequate dose.
- **Internal Target Volume (ITV):** The volume obtained by adding an internal margin to CTV to account for internal physiological motion and shape/position variation.
- **Planning Target Volume (PTV):** CTV (or ITV) plus setup margin to account for patient positioning and machine-related geometric uncertainties during treatment delivery.
- **Organ at Risk Volume (PRV / OAR volume):** A margin-expanded planning representation of an organ at risk, used to better protect normal tissues under motion and setup uncertainty.
- **Irradiated Volume:** The tissue volume receiving a clinically significant dose level (commonly a threshold relative to prescription dose), usually larger than treated volume.
- **Maximum Target Dose:** The highest absorbed dose within the target region that is considered clinically meaningful.
- **Minimum Target Dose:** The lowest absorbed dose found within the target region.
- **Mean Target Dose:** The arithmetic average dose in the target volume, typically calculated over sampled dose points/voxels.
- **Median Target Dose:** The central dose value within the target dose distribution (50th percentile).
- **Modal Target Dose:** The most frequently occurring dose value in the target dose distribution.
- **Hot Spots:** A hot spot is an area outside the target that receives a higher dose than the specified target dose. Like the maximum target dose, a hot spot is considered clinically meaningful only if it covers an area of at least 2 cm².

These definitions provide a standard framework for prescription, comparison, and reporting of treatment plans across institutions [2].

3.1.2 Treatment plan and geometry definition

After prescription, treatment planning begins with defining target volumes and critical organs using CT, MRI, or PET imaging. This 3D delineation process, known as contouring or segmentation, identifies structural borders which can be digitally represented as binary masks or triangular meshes. While manual or semi-automated thresholding techniques have traditionally been used, modern workflows increasingly

rely on automated segmentation tools to improve efficiency and consistency. These advanced tools encompass atlas-based segmentation—utilizing deformable image registration from a library of clinical cases—as well as artificial intelligence (AI) methods based on convolutional neural networks (CNNs). Despite these advancements, all automated segmentations necessitate thorough manual review and clinical editing to ensure precision before treatment delivery.

3.1.3 Creation of a treatment plan

3.1.4 Dose display and analysis tools

1. Isodose curves
2. Dose distribution in axial, sagittal, and coronal planes(3D dose distribution)
3. Dose-volume histograms (DVHs)

3.1.5 Intensity modulation

1. **Forward and Inverse Planning:** In *forward planning*, the treatment planner places beams into a radiotherapy treatment planning system that can deliver sufficient radiation to a tumor while both sparing critical organs and minimizing the dose to healthy tissue. The planner takes a trial and error approach while using past experience to alter beam parameters, such as beam weights and beam angles, to achieve an acceptable dose distribution. This is achievable when there is little geometric complexity with respect to adjacent critical organs or where doses delivered may be well below dose constraints for critical organs such as in palliative treatments. The treatment planner has the capability of modifying aperture shapes (such as defined by MLCs), beam weights, angles, and dose rates; such a large number of variables rapidly begins to exceed the capabilities of human comprehension. Hence, the concept of *inverse planning* was developed, in which a mathematical optimization of beam delivery parameters is performed to achieve a treatment plan that meets desired dosimetric goals. The required dose goals or objectives are typically provided by the operator in the form of dose threshold points on a DVH curve. These may be simple minimum and maximum doses to structures, or relative organ volumes receiving a given dose. In its simplest form, inverse planning can be used to determine optimal beam weights for a multi-field 3DCRT treatment plan. Inverse planning is typically used to optimize beam delivery components, such as MLC segment shapes and weights, 2D incident fluence (later converted to something deliverable by an MLC), gantry rotation speed, and dose rate.

2. **The rationale for intensity modulation:** The primary rationale for intensity modulation arises from the vital clinical need to highly conform the prescribed dose to complex target volumes, particularly when sensitive organs are adjacent to or enveloped by concave tumor surfaces (e.g., a tumor wrapping around the spinal cord).

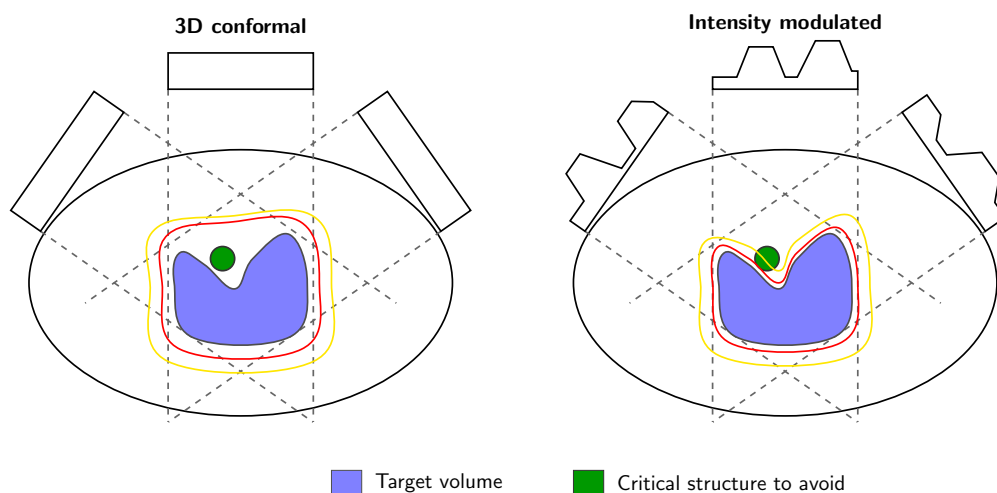


Figure 3: Conformity of dose to a target volume with a concave surface cannot be achieved via simple beam's-eye view planning with unmodulated coplanar beams (left). At right, intensity-modulated beams conform the dose distribution exactly to the target, successfully sparing the critical structure.

While conventional 3D conformal radiotherapy (3DCRT) shapes beam apertures using a 2D beam's-eye view (BEV), it fails to create conformal 3D dose distributions for hollow or cup-shaped targets because unmodulated uniform beams cannot selectively spare the nested critical structures. To achieve true volumetric conformity, non-uniform 2D beam intensity profiles must be generated. This typically involves a two-step planning process: determining suitable beam intensity maps through an inverse planning optimization algorithm, and then physically realizing these intensity profiles using computerized beam delivery mechanisms such as multileaf collimators (MLCs). Occasionally, these steps are combined via direct aperture optimization to overcome the mechanical limitations of the delivery device.

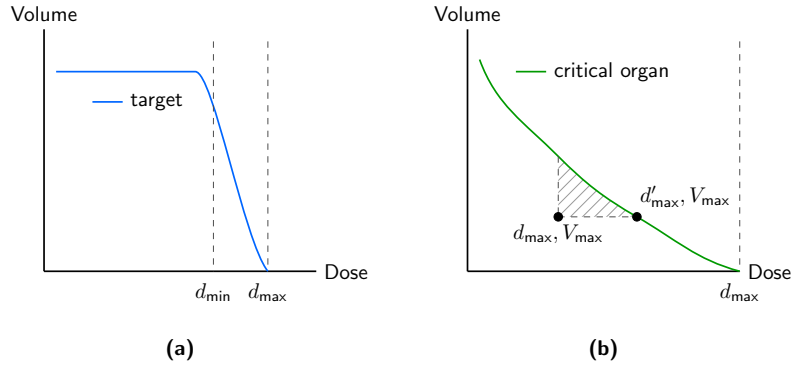


Figure 4: (a) Maximum and minimum target dose goals in terms of a DVH display. Any voxel with dose above the maximum and below the minimum will be penalized. (b) Maximum critical organ dose goal and a critical organ DVH goal. For the DVH goal we want the DVH curve to ideally reach that point. If the DVH passes underneath the point, that DVH goal is considered to be met and does not contribute to the optimization.

3. Simple optimization mathematics

Consider the schematic shown in Figure 5. If the matrix containing all beamlet weights is w and the matrix of calculated doses in the patient is d , the dose distribution is obtained from the matrix multiplication $d = Dw$ where D gives the change in d for a unit increment in each beam weight. Note that d and w are single-column matrices and D is a rectangular matrix with the same number of rows as d and the same number of columns as w has rows. In fact, d has as many rows as there are dose calculation points and w has as many rows as there are beamlets in the problem. Figure 5 shows a schematic of an intensity profile (the weights included in the w vector) and a dose distribution described by the d vector. An arrow is drawn (representing an element of the D matrix) indicating the influence of a beam element on a dose element.

Ideally, we would simply invert this matrix algebra expression and obtain w from $w = D^{-1}d$, but this is not a straightforward procedure for reasons such as those discussed above.

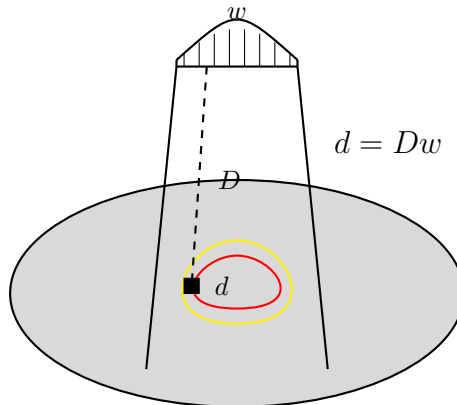


Figure 5: Schematic of an intensity profile (the weights included in the w vector) and a dose distribution described by the d vector. An arrow is drawn (representing an element of the D matrix) indicating the influence of a beam element on a dose element.

It is more instructive to write the dose calculation equation as a summation, where the dose in the i th voxel can be expressed as

$$d_i = \sum_{j=1}^M D_{ij} w_j. \quad (1)$$

Here j is the beamlet index and there are M beamlets over all beams used. Note that D_{ij} is the dose delivered to the i th voxel by the j th beamlet for a unit beam weight. A unit of beam weight may be a unit of beam fluence, energy fluence, opening density, exposure time, or any quantity that the D_{ij} matrix is normalized with respect to.

The Objective Function

An optimization algorithm needs an objective function to measure how far the result currently is from the desired solution. For a dose optimization problem, the objective function needs to express an overall voxel-by-voxel deviation from the desired dose, including the importance, or penalty, for each of those deviations. Deviations need not include a subtraction of desired dose from actual dose, but this is a useful and obvious measure. In fact, a quadratic objective function—including a weighted sum of the squared difference between desired and actual dose—is often used. As mentioned above, there may be several goal doses that apply to different voxels, and these goal doses act as the desired dose. If we denote the goal dose for the i th voxel as a prescript dose, d_i^{pres} , and the associated penalty for deviating from that goal as s_i , then the objective function can be defined as

$$F = \sum_{i=1}^N s_i (d_i - d_i^{pres})^2. \quad (2)$$

Here there are N voxels in which dose is to be considered throughout the dose matrix and corresponding to both targets and sensitive structures. The penalty values are typically user-defined by type and target/structure. If a voxel i is in a given structure and its dose value d_i exceeds the maximum dose goal for that structure, d_i^{pres} in Equation 2 will be the maximum dose goal for that structure. Similarly for structure voxels with dose beneath a minimum dose goal, d_i^{pres} is the minimum dose.

Iterative Optimization of Beamlet Weights

In order to drive the dose to a solution that better meets the pre-defined optimization goals, the most efficient tactic is to alter beamlet weights in some deterministic way. One such method is a so-called gradient technique, where the idea is to reduce the objective function value via calculated changes in the beamlet weights until its rate of change is zero. That is, the objective function is minimized. This concept is illustrated in Figure 6, where as iterations continue, the gradient of the objective function change decreases until it reaches a plateau and the user (or system) would cease the optimization.

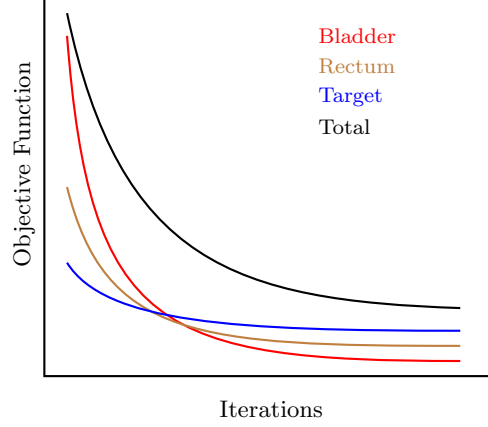


Figure 6: As the iterations progress, the objective function is reevaluated for each structure that has an objective assigned to it. The gradient descent process aims to reduce the difference between the desired dose objective and the actual dose value with each iteration, resulting in the objective function described in Equation 2 approaching zero as all objectives are met. Visualization of each individual objective function allows the user to determine which structure is contributing the most to the objective function, thus is farthest away from the desired dose.

According to the commonly used quasi-Newton gradient technique, the calculated changes in beamlet weight can be arrived at via the derivative and second derivative of the objective function with respect to that particular weight. The change in weight of beamlet j for a given iteration will be

$$\Delta w_j = -\alpha \left(\frac{dF}{dw_j} \middle/ \frac{d^2 F}{dw_j^2} \right). \quad (3)$$

The factor α is a damping term to ensure convergence to a solution. The proviso must be imposed that the adjusted weight cannot be less than zero (if it would become so, it is set to zero). Now using Equation (6.3), the change in weight becomes

$$\Delta w_j = -\alpha \frac{\sum_{i=1}^N s_i (d_i - d_i^{pres}) D_{ij}}{\sum_{i=1}^N s_i D_{ij}^2}. \quad (4)$$

Note that the weight change for beamlet j is essentially a weighted average of the dose-differences for all voxels influenced by that beamlet, where the weights in the average are dependent on the penalty value appropriate to that voxel and on the dose per unit beamlet weight in that voxel. Thus, as should intuitively be the case, the influence of a particular voxel’s dose-deviation on a given beamlet weight is larger when the dose per unit beamlet weight is higher for that voxel.

3.2 matRad Workflow

matRad is an open-source dose calculation and optimization toolkit developed entirely in MATLAB, explicitly designed for three-dimensional intensity-modulated radiation therapy (IMRT) research and educational purposes [1]. The software features a highly modular and sequential design that mirrors a clinical treatment planning environment. After importing patient anatomical models—either from external DICOM data or utilizing pre-loaded datasets like the CORT database—users can systematically compute and evaluate plans. The standard **matRad** workflow comprises four primary sequential processes:

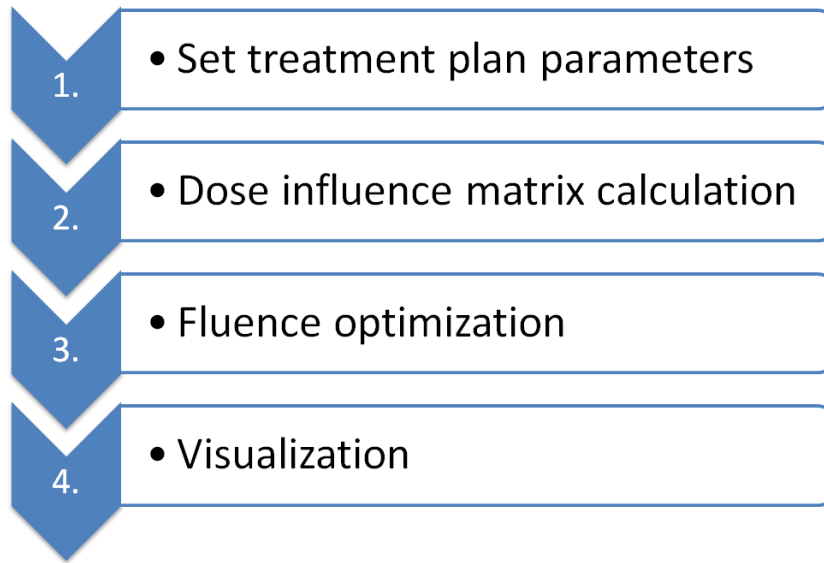


Figure 7: The modular workflow of `matRad` illustrating the sequential progression from plan parameterization to final visualization.

1. **Set Treatment Plan Parameters:** The workflow begins by initializing the dosimetric and geometric constraints. In this module, planners configure the desired irradiation modality (photons, scanned protons, or scanned carbon ions), define beam geometries including gantry and couch angles, establish grid resolutions, and delineate the requisite physical targets alongside proximate Organs at Risk (OARs) using the provided structural data.
2. **Dose Influence Matrix Calculation:** Once the spatial architecture is parameterized, `matRad` runs physical dose calculation algorithms to formulate the dose influence matrix (the D_{ij} matrix). This process computes the differential dose contributed by every individual non-zero beamlet or scanning pencil beam to each corresponding voxel within the volumetric CT grid. For scanned carbon ions, this module concomitantly calculates the physical dose and the associated Relative Biological Effectiveness (RBE).
3. **Fluence Optimization:** Following the matrix derivation, the system engages its central inverse planning engine. The fluence optimization module applies numerical optimization routines to resolve the ideal intensity weights of the multiple independent beamlets. By balancing customized objective functions and physiological penalties—maximizing conformality to the tumour volume while stringently minimizing dose to healthy tissue structures—the optimizer converges upon a clinically acceptable fluence distribution.
4. **Visualization:** To conclude the workflow, the visualization engine structurally renders the outcome. Treatment planners are provided graphical user interfaces (GUI) to visually interpret the 3D dose distributions overlaid on axial, coronal, and sagittal CT slices. Furthermore, the module displays quantitative plan evaluations, such as Dose-Volume Histograms (DVHs), permitting a thorough and rigorous assessment of the resultant treatment topology [1].

4 Observation

5 Conclusion

References

- [1] H.-P. Wieser, E. Cisternas, N. Wahl, S. Ulrich, A. Stadler, H. Mescher, L.-R. Müller, T. Klinge, H. Gabrys, L. Burigo, A. Mairani, S. Ecker, B. Ackermann, M. Ellerbrock, K. Parodi, O. Jäkel, and M. Bangert, “Development of the open-source dose calculation and optimization toolkit `matRad`,” *Medical Physics*, vol. 44, no. 6, pp. 2556–2568, 2017.



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